

AI-Powered Citizen Complaint Management System

Problem Statement

Many municipalities and local government organizations still rely on manual methods to manage citizen complaints. Citizens report issues such as water leakage, garbage accumulation, road damage, street light failures, drainage blockages, and public park maintenance through phone calls, paper forms, or direct visits to government offices. These complaints are manually forwarded to different departments, making it difficult to monitor their progress and ensure timely resolution.

This manual complaint handling process leads to several challenges:

- Manual submission and registration of citizen complaints.
- Difficulty in identifying the appropriate department for complaint resolution.
- Lack of automated complaint prioritization.
- No centralized complaint tracking system.
- Delays in assigning complaints to responsible departments.
- Increased paperwork and administrative effort.
- Lack of automated notifications for complaint updates.
- Absence of AI-assisted complaint summarization and recommendations.
- Limited reporting and dashboard capabilities for monitoring complaint trends.
- Higher possibility of human errors during complaint processing.
- These limitations reduce operational efficiency, delay complaint resolution, and negatively impact citizen satisfaction.

Objective

The primary objective of this project is to develop an **AI-Powered Citizen Complaint Management System** using the ServiceNow platform to automate the complete complaint management lifecycle.

Key Objectives:

- Design a centralized application for citizen complaint management.
- Develop custom tables to manage citizen complaints and departments.
- Automatically assign complaint priority using Client Scripts.
- Automatically assign complaints to the appropriate department using Business Rules.
- Generate AI-powered complaint summaries.
- Generate AI-based recommendations for complaint resolution.
- Automate notifications using Flow Designer.
- Implement role-based security using ACLs.
- Create reports and dashboards for complaint analysis.
- Develop Knowledge Articles for common complaint resolution procedures.
- Demonstrate ServiceNow AI concepts such as AI Summary, AI Recommendation, AI Routing, AI Search (requested activation), and Now Assist overview.
- Improve complaint handling efficiency while reducing manual intervention.
- Automatically maintain complaint records throughout the complaint lifecycle.
- Provide secure access to complaint information using Roles and ACLs.
- Utilize Flow Designer to automate notifications and complaint updates with minimal scripting.

Phase 1 — Requirement Analysis & Planning

Activity: 1 Business Objective

The purpose of the **AI-Powered Citizen Complaint Management System** is to digitize and automate the complaint management process for municipal organizations using the ServiceNow platform.

The application improves operational efficiency, reduces manual work, and enhances citizen satisfaction by providing a transparent and automated complaint management process.

Business Problems Identified

- Complaints are not assigned automatically to departments.
- Lack of centralized complaint tracking.
- Delays in complaint resolution.
- No AI-assisted complaint recommendations.
- Manual priority assignment.
- Lack of automated notifications.
- No centralized reporting system.
- Limited visibility into complaint status.
- Increased administrative workload.
- No centralized reporting system

Business Goals

- Automatically assign complaint priority.
- Automatically route complaints to the appropriate department.
- Generate AI-based complaint summaries.
- Generate AI recommendations for complaint resolution.

- Automate complaint notifications.
- Improve transparency for citizens and municipal staff.
- Generate reports and dashboards for monitoring complaint trends.
- Demonstrate ServiceNow AI capabilities through AI Summary, AI Recommendation, and AI Routing
- Demonstrate AI-assisted summarization using the Now Assist concept.

Activity: 2 Functional Scope

This section defines what the application will do (and what it will not).

In Scope (What We Will Build)

Citizen Management

- Citizens can log in and register a complaint.
- Citizens can upload an image as evidence.
- Citizens can track complaint status in real time.

Department Management

The application automatically assigns complaints to the appropriate department based on the selected complaint category.

departments included:

- Water Department
- Sanitation Department
- Electrical Department
- Roads Department
- Parks Department

Complaint Cycle

The complaint moves through the following stages:

Complaint Created



Priority Assigned



Department Assigned



AI Summary Generated



AI Recommendation Generated



Notification Sent



Complaint Resolved





Complaint Closed

Complaint Form

Citizens submit complaints using the Citizen Complaint form.

The form captures:

- Citizen Name
- Complaint Title
- Category
- Description
- Location
- Priority
- Status
- Assigned Department
- AI Summary
- AI Recommendation
- Resolution

Automation

The application uses **Flow Designer** to automate:

- Complaint Notifications
- Complaint Updates
- Email Notifications
- Workflow Automation

Business rule

Business Rules are implemented for:

- Automatic Department Assignment
- AI Recommendation Generation
- AI Summary Generation

Minimal server-side scripting is used to automate complaint processing.

Client Scripts

Client Scripts are used for:

- Automatic Priority Assignment based on Complaint Category.
- Improve user experience during complaint creation.

UI Policy

The UI Policy automatically displays the **Resolution** field only when the complaint status changes to **Resolved**.

AI Integration

The application demonstrates ServiceNow AI concepts through:

- AI Complaint Summary
- AI Recommendation Engine
- AI-Based Department Routing
- Knowledge Management
- AI Search (Activation Requested)
- Now Assist Overview

Activity: 3 Stakeholder Mapping

Understanding who uses the system and how.

Stakeholder	Role	Responsibilities
Citizen	Basic user	Register complaint, upload evidence, track status
Officer	Department staff	View assigned complaints, resolve complaints
Manager	Department oversight	Monitor dashboard, review reports
Admin	System configuration	Configure application, roles, and security
AI Agent	Automated assistant	Summarize, predict category, suggest action, reply to citizen

Activity: 4 Execution Roadmap (Milestones)

This roadmap shows how the project will be built step-by-step.

Milestone 1 — Application Setup

- Create Scoped Application.
- Configure application settings.
- Create custom tables.

Milestone 2 — Data Model

- Create custom table:
 - Citizen Complaint
 - Department

Milestone 3 — Form Design

Configure:

- Complaint Form

- Choice Fields
- Reference Fields
- Related List

Milestone 4 — Client Scripts

Implement:

- Automatic Priority Assignment.

Milestone 5 — Business Rules

Implement:

- Department Assignment.
- AI Recommendation.
- AI Summary.

Milestone 6 —Flow Designer

- Develop automation for:
- Complaint Notifications.
- Complaint Updates.

Milestone 7 — Security (ACLs)

- Read, Write, Delete ACLs.
- Role-based and requester-based access.

Milestone 8 — Reports

Create:

- Complaints by Category
- Complaints by Status

- Complaints by Priority
- Complaints by Department
- AI Routed Complaints

Milestone 9 — Dashboard

Develop the **Citizen Complaint AI Dashboard** to monitor complaint statistics and AI-enabled complaint routing.

Milestone 10 — Knowledge Management

Create Knowledge Articles for:

- Water Leakage Resolution
- Street Light Maintenance
- Garbage Collection Procedure
- Drainage Blockage Resolution
- Road Damage Repair

PHASE 2 — Backend Development & Configuration

In this phase, we design and configure the backend foundation of the **AI-Powered Citizen Complaint Management System** using the ServiceNow platform.

This phase includes creating custom tables, configuring fields, relationships, forms, security roles, and implementing backend logic required to automate the complete citizen complaint management process.

The application is developed as a **Scoped Application** using ServiceNow Studio to ensure modularity, maintainability, and security.

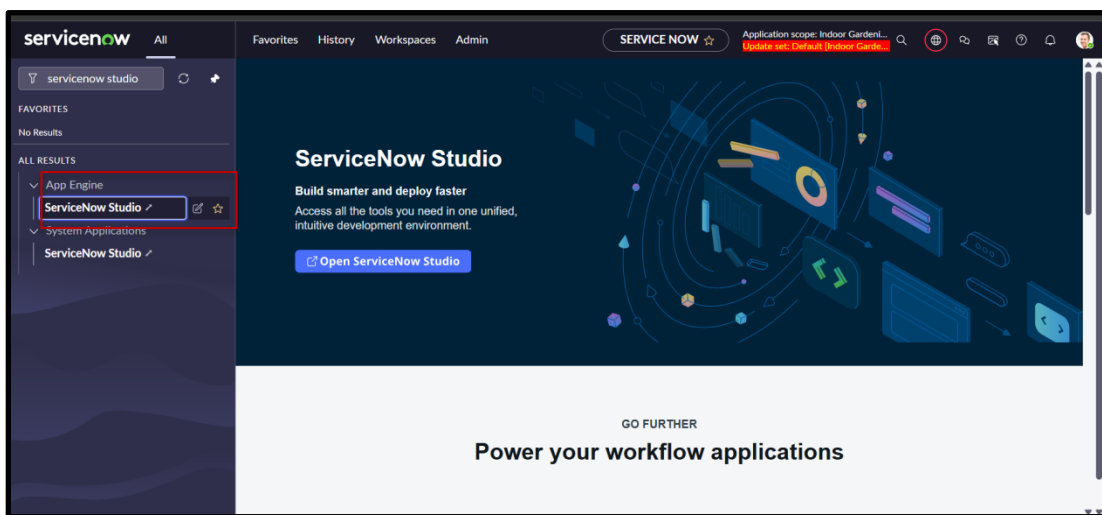
Data Architecture

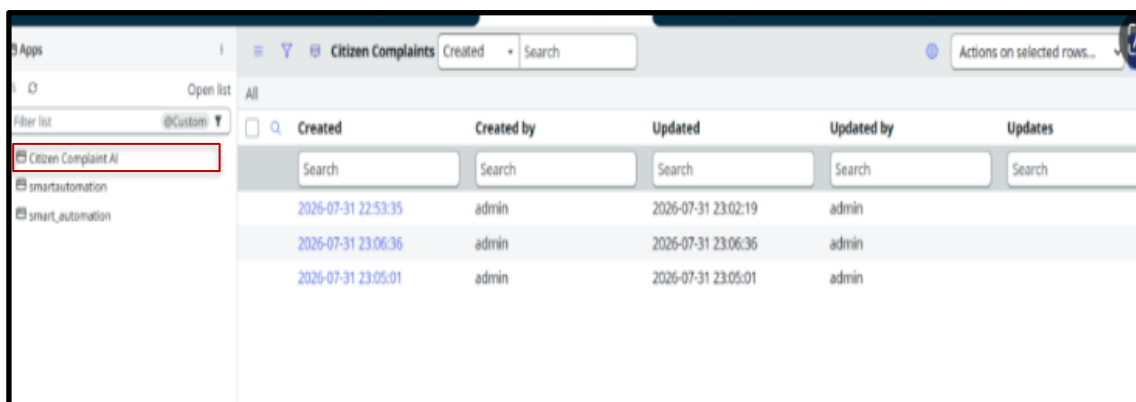
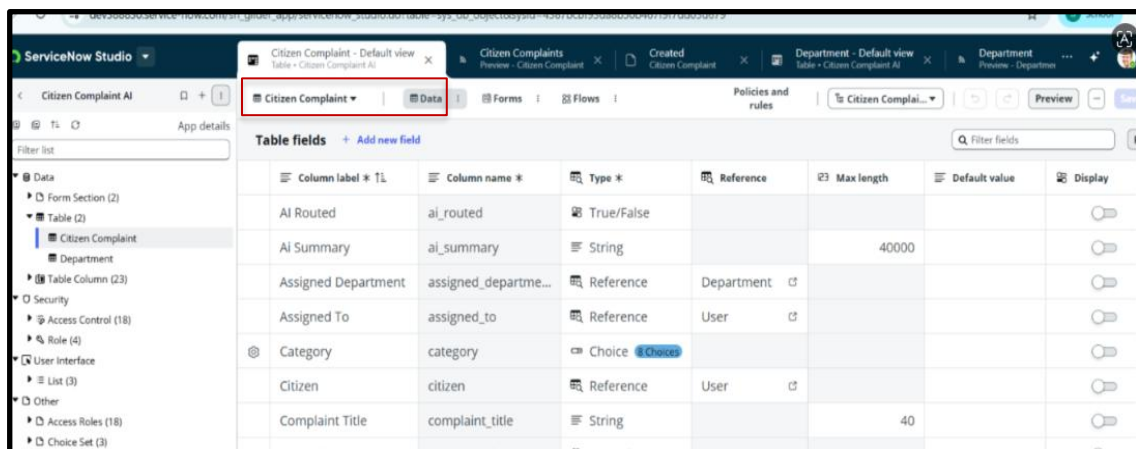
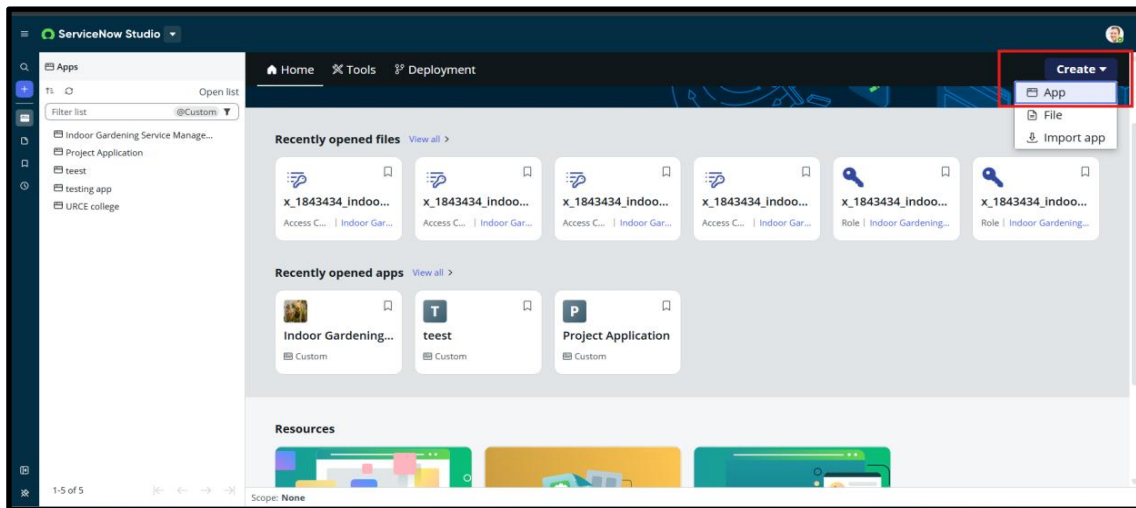
Creation of Scoped Application

- The application was developed using **ServiceNow Studio** as a Scoped Application.
- **Steps to Create the Application**

- Open the ServiceNow Personal Developer Instance (PDI).
- Search for **Studio** in the Application Navigator.
- Click **Create Application**.
- Enter the application details.

Property	Value
Application Name	AI-Powered Citizen Complaint Management System
Scope	x_1972253_citize_0 (<i>Use your actual scope name</i>)
Platform	ServiceNow
Development Type	Scoped Application
- Click **Create**.
- Configure the application modules.
- Create custom tables.
- Configure forms and related lists.
- Configure Business Rules, Client Scripts, UI Policies, Flow Designer, Notifications, Reports, Dashboard, and Knowledge Articles..
- Now see the following steps as follows in picture view.





Timestamp	User	Complaint Details
2026-07-31 23:30:57	admin	2026-07-31 23:34:05 admin 1 (empty)
2026-08-01 05:36:16	admin	2026-08-01 05:36:16 admin 0 1a1f43f931e8b50b467f9f7dd03d6ba
2026-08-01 00:29:43	admin	2026-08-01 00:29:52 admin 1 1a1f43f931e8b50b467f9f7dd03d6ba
2026-07-31 22:53:35	admin	2026-07-31 23:02:19 admin 1 058f74ff931e8b50b467f9f7dd03d6ba
2026-07-31 23:16:07	admin	2026-07-31 23:16:07 admin 0 (empty)

Creation of Roles

Three custom roles were created to provide secure access to the application.

Role	Purpose
Citizen	Submit and view complaints

Department Officer Update complaints assigned to their department

Role	Purpose
Administrator	Full access to configure and manage the application

These roles ensure that users can only access records according to their responsibilities.

Let's add permissions to your table.

Create roles and define how much control each role has of this table. Note: at least one role needs to have 'read' access for you to 'preview' the data in your table.

Type	Role Name	Description	All	Create	Read	Write	Delete
Existing	x_1972253_citize_0.admin	Application Administrator	☒	☒	☒	☒	☒
Existing	x_1972253_citize_0.citizen	Citizens can create and vi...	☐	☐	☐	☐	☐
Existing	x_1972253_citize_0.mana...	Managers can view dashb...	☐	☐	☐	☐	☐
Existing	x_1972253_citize_0.officer	Department officers can ...	☐	☐	☐	☐	☐

PHASE 3 — Application Development & Execution

In this phase, we build and execute the Ai powered citizen complaint Management application using ServiceNow's low-code features such as Flow Designer, ACLs, Business Rules, and Client Scripts.

Activity: 3.1 table configuration

Key Fields

Field	Type
Citizen Name	String
Phone	String
Email	Email

Field	Type
Location	String
Complaint Type / Category	Choice
Description	String (multi-line)
Image	Attachment
Priority	Choice
Status	Choice
AI Summary	String (multi-line)
Predicted Category	String
Assignment Group / Department	Reference
Resolution	String (multi-line)
Created Date	Date/Time
Resolved Date	Date/Time
Knowledge Generated	True/False

The screenshot shows the ServiceNow Studio interface for configuring a table named 'Citizen Complaint AI'. The 'Table fields' section is active, displaying a list of fields with their respective types and configurations. A red box highlights the 'Table fields' section.

Column label * TL	Column name *	Type *	Reference	Max length	Default value	Display
AI Routed	ai_routed	True/False				
AI Summary	ai_summary	String		40000		
Assigned Department	assigned_departme...	Reference	Department			
Assigned To	assigned_to	Reference	User			
Category	category	Choice				
Citizen	citizen	Reference	User			
Complaint Title	complaint_title	String		40		
Created	sys_created_on	Date/Time				
Created by	sys_created_by	String		40		

Citizen Complaint
New record

Landmark: gachibowli

Citizen: Alejandra Prenatt

Assigned To: Alissa Mountjoy

Location: madhapur

Description: street light

Priority: High

Resolution:

Category: Drainage

Assigned Department:

Status: Resolved

Complaint Title: street light

Submit

Department Table

Field	Type
Department Name	String
Manager	Reference (User)
Email	Email
Location	String

ServiceNow Studio

Citizen Complaint AI

Department

Table fields

Column label *	Column name *	Type *	Reference	Max length	Default value	Display
Active	active	True/False				☐
Created	sys_created_on	Date/Time				☐
Created by	sys_created_by	String		40		☐
Department Code	department_code	String		40		☐
Department Head	department_head	Reference	User			☐
Department Name	department_name	String		1000		☐
Description	description	String		40000		☐
Email	email	String		40		☐
Location	location	String		200		☐

The screenshot displays the 'Citizen Complaint AI' application interface. On the left is a sidebar with a 'Filter list' and a tree view containing categories like Data, Form Section, Table, Citizen Complaint, Department, Table Column, Security, Access Control, Role, Server Development, User Interface, List, Other, and Access Roles. The main area shows a table titled 'Departments' with a 'Created' filter. The table has columns for 'Created', 'Created by', 'Updated', 'Updated by', 'Updates', 'Department Name', 'Department Code', and 'Location'. A red box highlights a subset of the data rows.

Created	Created by	Updated	Updated by	Updates	Department Name	Department Code	Location
2026-07-31 22:48:45	admin	2026-07-31 22:48:45	admin	0	Parks Department	EDD05	Zone D
2026-07-31 22:47:17	admin	2026-07-31 22:47:17	admin	0	Water Department	WD01	Zone Awat
2026-07-31 22:49:40	admin	2026-07-31 22:49:40	admin	0	sanitary department	EDD76	Zone E
2026-07-31 22:48:10	admin	2026-07-31 22:48:10	admin	0	Electrical Department	EL002	Zone C
2026-07-31 22:40:43	admin	2026-07-31 22:46:13	admin	2	Electrical Department	WD003	Zone C

Activity: 3.2 Form Configuration

The Citizen Complaint form was configured to provide a user-friendly interface for complaint registration and management.

The form includes:

- Complaint Details
- Department Information
- AI Features
- Resolution Information

The layout was customized to improve usability and simplify complaint management.

The screenshot shows the 'Personalize Form' configuration screen. It features a title bar with a search icon and a list of 'Form Fields' with checkboxes. A red box highlights the list of fields.

Form Fields
☒ AI Routed
☒ AI Summary
☒ Assigned Department
☒ Assigned To
☒ Category
☒ Citizen
☒ Complaint Title
☒ Description
☒ Landmark
☒ Location
☒ Priority
☒ Resolution
☒ Status

Activity: 3.3 Client Script

The Client Script automatically assigns the complaint priority based on the selected complaint category.

Example

- Water Leakage → High
- Garbage → Medium
- Street Light → Medium
- Road Damage → High
- Parks → Low

This reduces manual effort and ensures consistent priority assignment.

Steps

- Navigate to **Client Scripts**.
- Create a new Client Script.
- Select the Citizen Complaint table.
- Choose **on Change**.
- Select **Category** field.
- Write the JavaScript logic for automatic priority assignment.

Activity: 3.4 Business Rule

Purpose of Business Rule

The Business Rule automatically assigns the complaint to the appropriate department based on the selected complaint category.

Department Routing

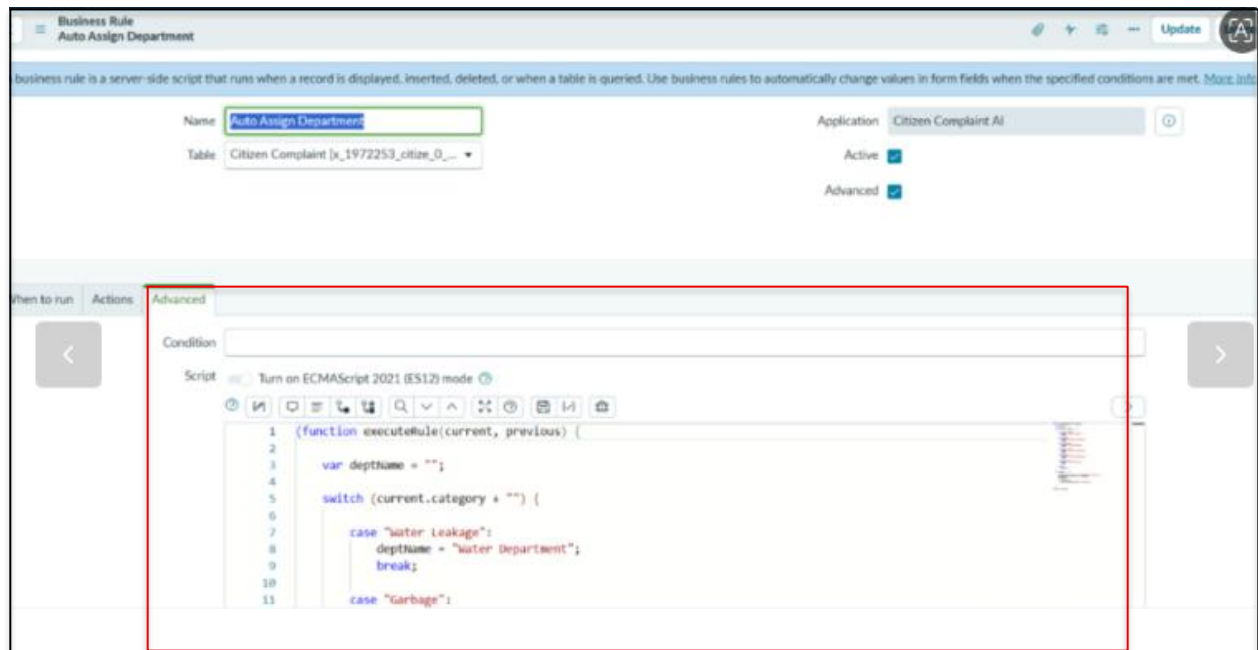
Category	Department
Water Leakage	Water Department
Garbage	Sanitation Department
Street Light	Electrical Department

Category	Department
Road Damage	Roads Department
Parks	Parks Department

The Business Rule executes before inserting or updating the complaint record.

This eliminates manual department assignment.

Auto Assignment — map complaint category to department



Activity 3.5 – UI Policy

The UI Policy controls the visibility of the **Resolution** field.

LOGIC

If
 Status = **Resolved**
 Then
 Show Resolution Field
 Otherwise
 Hide Resolution Field.
 This improves the user interface by displaying the Resolution field only when required.

Steps

- Navigate to **UI Policies**.
- Create a new UI Policy.
- Select the Citizen Complaint table.
- Set Condition:

Status = Resolved

- Create UI Policy Action.

Set

Resolution

Visible = True

Mandatory = False

The screenshot shows the 'UI Policy - New Record' page in ServiceNow. The page is titled 'UI Policy - New Record' and has a search bar. The 'Table' dropdown is set to 'Citizen Complaint [x_1972253_citizen_0...]' and the 'Application' is 'Citizen Complaint AI'. The 'Active' checkbox is checked. The 'Short description' field contains 'Displays the Resolution field only when the complaint status is Resolved.' and the 'Order' field is set to 100. The 'When to Apply' tab is selected, showing the 'Conditions' section. The condition is 'Status is Resolved' with 'AND' and 'OR' options. The 'Global' checkbox is checked. The 'On load' checkbox is also checked. The 'Reverse the effects of the UI policy actions when the Conditions evaluate to false' checkbox is checked. The 'Inherit' checkbox is unchecked.

New record

UI policy actions specify exactly what actions to take on a specified field. The conditions specified in the UI policy determine when these actions are triggered. [More Info](#)

UI policy
Displays the Resolution field only when t

Application
Citizen Complaint AI

Table
Citizen Complaint
[x_1972253_citize_0_citizen_complaint]

* Field name
Resolution

Mandatory
True

Visible
True

Read only
Leave alone

Clear the field value

dev388830.service-now.com/now/nav/ui/classic/params/target/sys...

servicenow All UI Policy - Dis...

UI Policy
Displays the Resolution field only when the complaint status is Resolved.

Update Delete

Related Links
Convert this to Data Policy
Advanced view
Run Point Scan

UI Policy Actions UI Policy Related List Actions

for text Search

UI policy = Displays the Resolution field only when the complaint status is Resolved.

Field name	Mandatory	Visible	Read only
			

Run scan on the application

Activity 3.6 – AI Recommendation

Example

Water Leakage

AI Recommendation:

Assign immediately to Water Department and inspect the damaged pipeline.

Garbage

AI Recommendation:

Schedule waste collection and sanitation inspection.

Road Damage

AI Recommendation:

Forward complaint to Roads Department for inspection and repair.

These recommendations assist municipal officers during complaint resolution.

	Created	Created by	Updated	Updated by	Updates	Assigned Department	AI Summary	AI Routed	AI Recommendation
	Search	Search	Search	Search	Search	Search	Search	Search	Search
☐	2026-07-31 23:40:29	admin	2026-07-31 23:40:29	admin	0	cb5ff43f931e8b50b467f9f7dd03d63f		false	
☐	2026-08-01 00:02:03	admin	2026-08-01 00:02:03	admin	0	1cbf38f931e8b50b467f9f7dd03d6ab		false	
	2026-07-31 23:30:57	admin	2026-07-31 23:34:05	admin	1	(empty)		false	
	2026-08-01 05:41:11	admin	2026-08-01 05:41:11	admin	0	1a1fb43f931e8b50b467f9f7dd03d6ba	Complaint: water contaminated. Category:...	true	
	2026-08-01 08:17:45	admin	2026-08-01 08:17:45	admin	0	1a1fb43f931e8b50b467f9f7dd03d6ba	Complaint: water leak related. Category:...	true	Review complaint manually and assign the...
	2026-08-01 05:36:16	admin	2026-08-01 05:36:16	admin	0	1a1fb43f931e8b50b467f9f7dd03d6ba	Complaint: Water Leakage Near School. Ca...	false	

Activity 3.7 – AI Summary

Purpose

The AI Summary field automatically stores a concise summary of the complaint.

Sample AI Summary

Citizen reported a water leakage issue near the public park. Complaint categorized as Water Leakage and routed to the Water Department for immediate inspection.

The AI Summary provides a quick overview of the complaint without reading the complete description.

	Created	Created by	Updated	Updated by	Updates	Assigned Department	AI Summary
	Search	Search	Search	Search	Search	Search	Search
☐	2026-07-31 23:40:29	admin	2026-07-31 23:40:29	admin	0	cb5ff43f931e8b50b467f9f7dd03d63f	
☐	2026-08-01 00:02:03	admin	2026-08-01 00:02:03	admin	0	1cbf38f931e8b50b467f9f7dd03d6ab	
	2026-07-31 23:30:57	admin	2026-07-31 23:34:05	admin	1	(empty)	
	2026-08-01 05:41:11	admin	2026-08-01 05:41:11	admin	0	1a1fb43f931e8b50b467f9f7dd03d6ba	Complaint: water contaminated. Category:...
	2026-08-01 08:17:45	admin	2026-08-01 08:17:45	admin	0	1a1fb43f931e8b50b467f9f7dd03d6ba	Complaint: water leak related. Category:...
	2026-08-01 05:36:16	admin	2026-08-01 05:36:16	admin	0	1a1fb43f931e8b50b467f9f7dd03d6ba	Complaint: Water Leakage Near School. Ca...

Activity 3.8 – Flow Designer

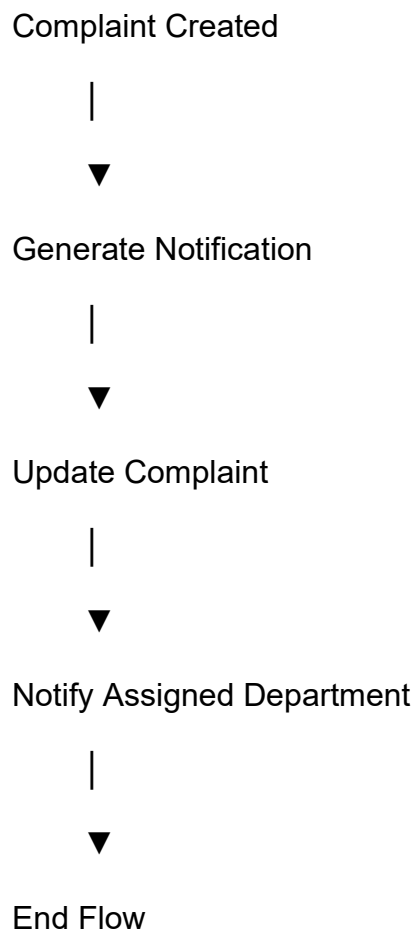
Purpose

Flow Designer automates the complaint workflow without heavy coding.

Trigger

When a new complaint record is created.

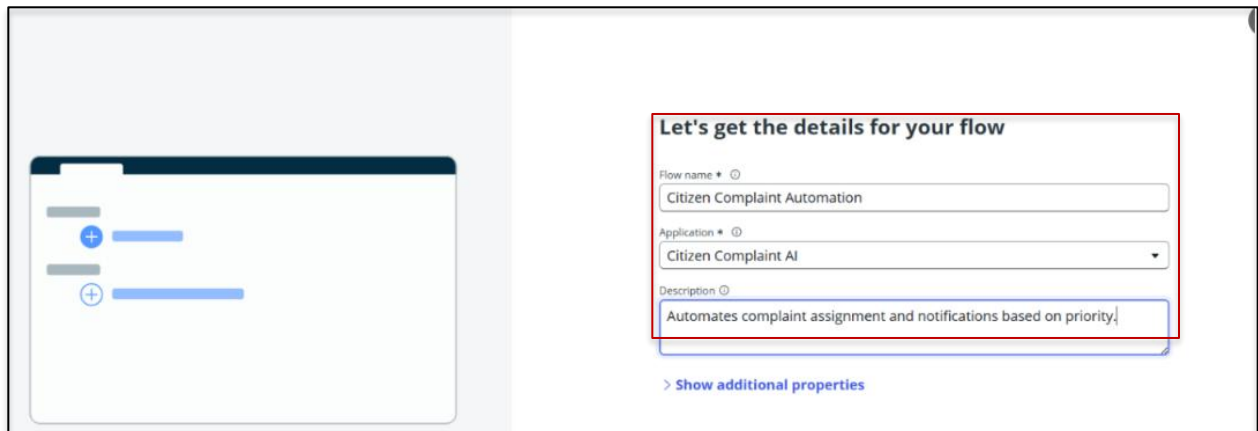
Flow Steps



Benefits

- Automated workflow
- Reduced manual work

- Faster complaint processing
- Improved communication



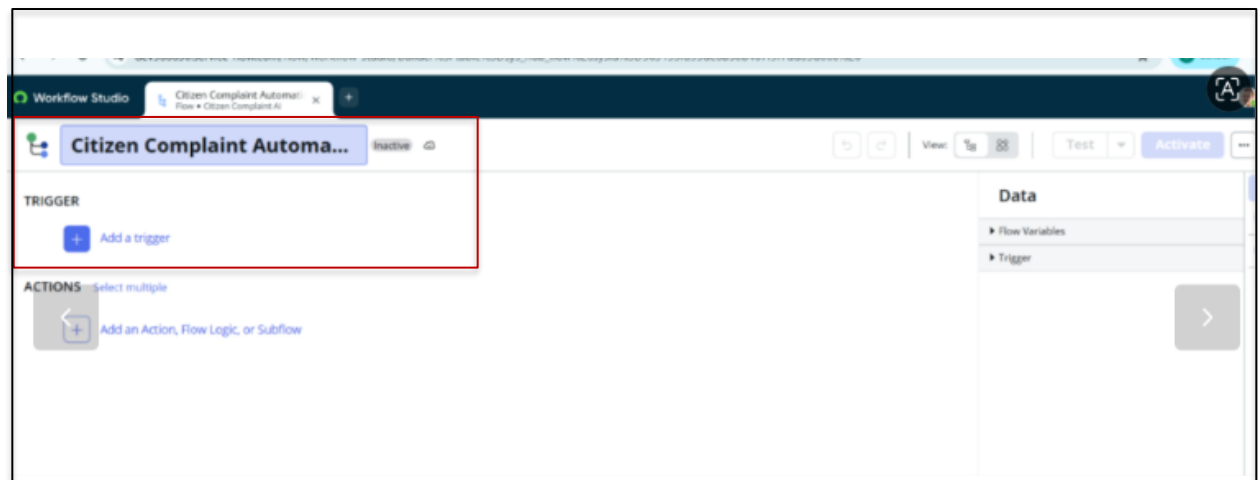
Let's get the details for your flow

Flow name * 
 Citizen Complaint Automation

Application * 
 Citizen Complaint AI

Description 
 Automates complaint assignment and notifications based on priority

[Show additional properties](#)



Workflow Studio Citizen Complaint Automation x +

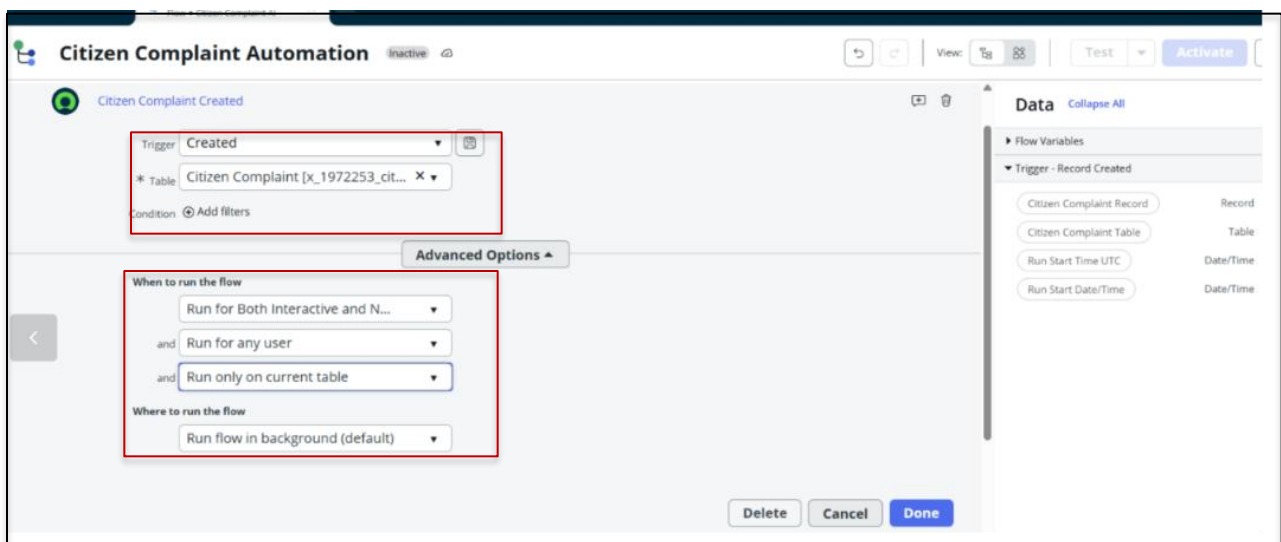
Citizen Complaint Automa... Inactive 

TRIGGER  Add a trigger

ACTIONS  Add an Action, Flow Logic, or Subflow

Data

- Flow Variables
- Trigger



Citizen Complaint Automation Inactive 

Citizen Complaint Created

Trigger: Created 

* Table: Citizen Complaint [x_1972253_cit... x v]

Condition:  Add filters

Advanced Options

When to run the flow

- Run for Both Interactive and N... 
- and Run for any user 
- and Run only on current table 

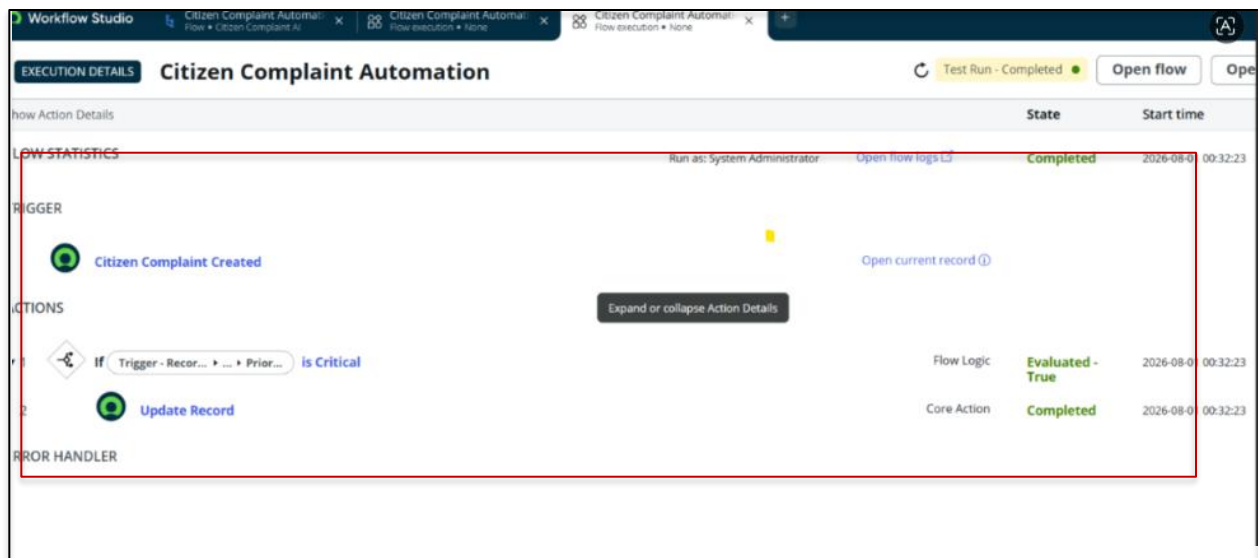
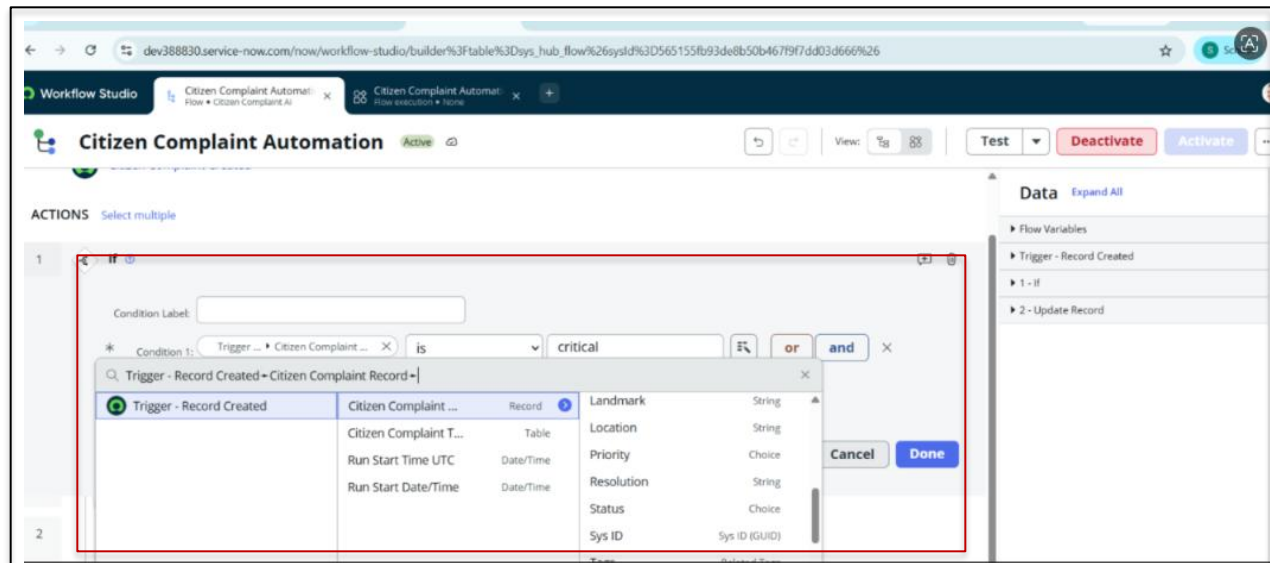
Where to run the flow

- Run flow in background (default) 

Data 

- Flow Variables
- Trigger - Record Created
 - Citizen Complaint Record Record
 - Citizen Complaint Table Table
 - Run Start Time UTC Date/Time
 - Run Start Date/Time Date/Time

Delete Cancel Done



Activity 3.9 – Notification

The application automatically sends an email notification whenever a complaint is created.

The notification contains:

- Complaint Number
- Citizen Name
- Category

- Assigned Department
- Priority
- Status

This ensures that responsible departments receive complaint information immediately.

Notification [sysevent_email_action]
New record

Mandatory ☐

* Category: Uncategorized

A digest combines and summarises multiple updates for a single or multiple target records within a set time interval.

Allow Digest ☐

When to send | Who will receive | What it will contain

Notifications can be sent (if the specified **Conditions** are met) under one of the following circumstances:

- A record is **Inserted** or **Updated** into the **Table** specified above
- The specified event is fired
- Via a Flow Action

Send when: Record inserted or updated

Inserted ☐

Updated ☐

Conditions: Add Filter Condition Add OR Clause

-- choose field -- -- oper -- -- value --

When to send | Who will receive | What it will contain

Notifications can be sent to specific **Users** and **Groups** or to **User/Groups in fields** on the record that generated this notification.

Users

Users/Groups in fields

Groups

Subscribable ☐

When to send | Who will receive | **What it will contain**

When using an Email Template, the Subject and Message content is picked from that template unless a different Subject or Message content is specified in this form.

Email template
Unsubscribe and Preferences

Subject

Message HTML

Select variables:

B I U ↶ ↷ ↵ Verdana 8pt

A ↕ ↗ ↘ ↙ ↚ ↛ ↜ ↝ ↞ ↠ ↡ ↢ ↣ ↤ ↥ ↦ ↧ ↨ ↩ ↪ ↫ ↬ ↭ ↮ ↯ ↰ ↱ ↲ ↳ ↴ ↵ ↶ ↷ ↸ ↹ ↺ ↻ ↼ ↽ ↾ ↿ ⇀ ⇁ ⇂ ⇃ ⇄ ⇅ ⇆ ⇇ ⇈ ⇉ ⇊ ⇋ ⇌ ⇍ ⇎ ⇏ ⇐ ⇑ ⇒ ⇓ ⇔ ⇕ ⇖ ⇗ ⇘ ⇙ ⇚ ⇛ ⇜ ⇝ ⇞ ⇟ ⇠ ⇡ ⇢ ⇣ ⇤ ⇥ ⇦ ⇧ ⇨ ⇩ ⇪ ⇫ ⇬ ⇭ ⇮ ⇯ ⇰ ⇱ ⇲ ⇳ ⇴ ⇵ ⇶ ⇷ ⇸ ⇹ ⇺ ⇻ ⇼ ⇽ ⇾ ⇿ ⇿ ⇿ ⇿ ⇿

Notifications Table Search

iii > Name starts with Complaint Assigned Notification

Name	Active	Table	Category	Email template	Conditions	Description	Users/Groups in fields	Users
Complain Search	Search	Search	Search	Search	Search	Search	Search	Search
Complaint Assigned Notification	true	Citizen Complaint [x_1972253_citize_0_citizen_complaint]	Uncategorized	Unsubscribe and Preferences	assigned_departmentISNOTEMPTY*EQ		assigned_to	
Complaint Assigned Notification	true	Citizen Complaint [x_1972253_citize_0_citizen_complaint]	Uncategorized	Unsubscribe and Preferences	assigned_departmentISNOTEMPTY*EQ		assigned_to	
Complaint Assigned Notification	true	Citizen Complaint [x_1972253_citize_0_citizen_complaint]	Uncategorized	Unsubscribe and Preferences	assigned_departmentISEMPTY*EQ		assigned_to	
Complaint Assigned Notification	true	Citizen Complaint [x_1972253_citize_0_citizen_complaint]	Uncategorized	Unsubscribe and Preferences	assigned_departmentISNOTEMPTY*EQ			System Administrator, bhavanisuravaram@g

PHASE 4 — Testing, Validation & Security

Area	Validation Result
Complaint Registration	Working
Auto Priority (Business Rule)	Applied
Client Script Validation	Working
Flow Designer Automation	Triggered
Now Assist Summary	Generated
Predictive Intelligence	Trained &Predicting
ACL Security	Enforced

Activity: 4.1 Testing Scenario

Scenario 1 – Complaint Creation

Test Objective

Verify that a citizen can successfully submit a complaint.

Expected Result

- Complaint record is created.
- Complaint Number is generated.
- Status is set to resolved.

Actual Result

Complaint created successfully.

The screenshot shows a mobile application interface for submitting a citizen complaint. The form is titled "Citizen Complaint" and "Created 2024-07-31 23:16:07". It contains the following fields and values:

- Landmark: near my village
- Citizen: Alissa Mountjoy
- Assigned To: Amelia Caputo
- Location: old monk
- Description: it related to water leak
- Priority: Medium
- Category: Water Leakage
- Assigned Department: (empty)
- Status: Resolved (highlighted with a red box)
- Complaint Title: water leaked for latest

At the bottom of the form, there are "Update" and "Delete" buttons.

Scenario 2 – Client Script Validation

Test Objective

Verify automatic priority assignment.

Example

Category = Water Leakage

Expected Priority = High

Actual Result

Priority assigned automatically

Apps

Open list

filter list @Custom

Citizen Complaint AI

smartautomation

smart_automation

Citizen Complaint
Created 2026-07-31 22:53:35

Landmark Government School

Citizen Abel Tuter

Assigned To Abraham Lincoln

Location MG Road

Description garage is leakign and not done cleanly

Priority High

AI Routed

AI Summary

Resolution

Category Garbage

Assigned Department Created 2026-07-31 22:48:45

Status New

Complaint Title Water leakage near school

Update

Delete

Scenario 3 – Business Rule Validation

Test Objective

Verify automatic department assignment.

Example

Category = Garbage

Expected Department

Sanitation Department

Actual Result

Department assigned successfully.

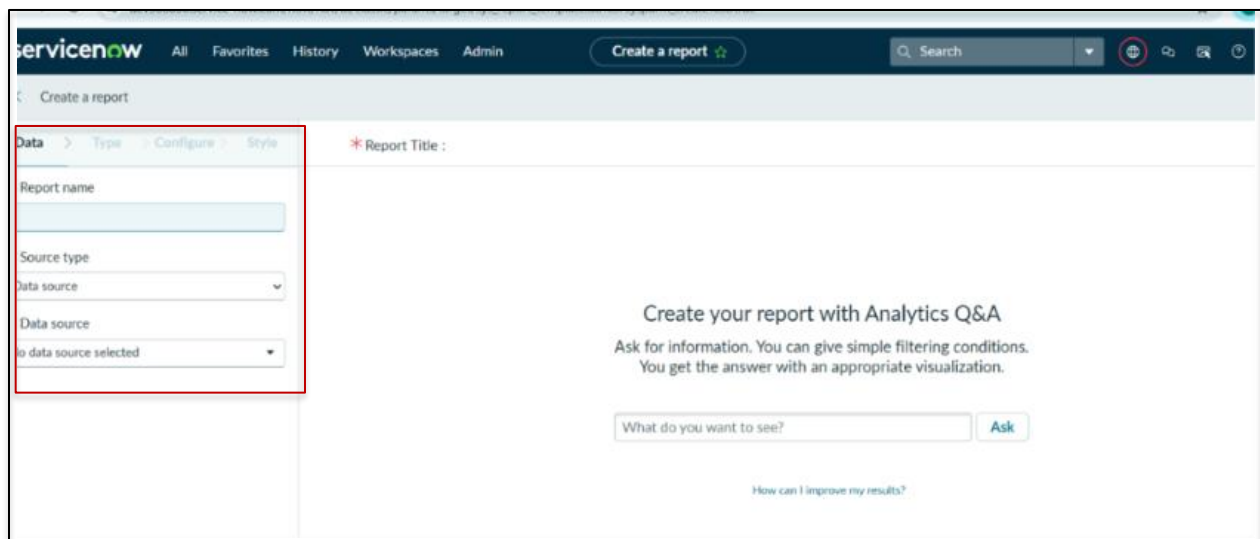
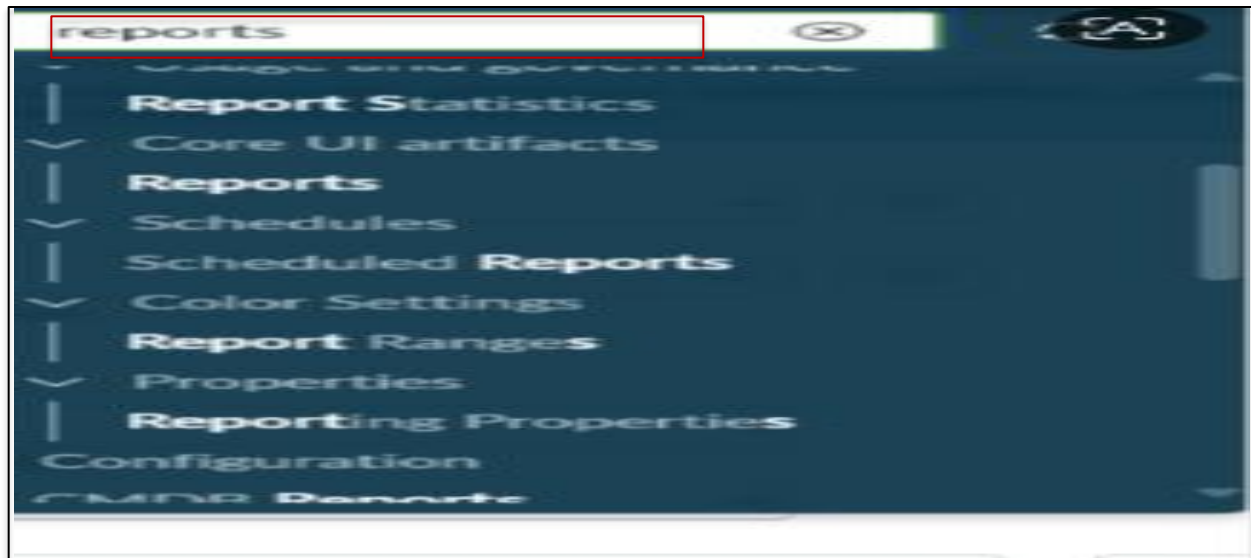
Created	Created by	Updated	Updated by	Updates	Assigned Department
Search	Search	Search	Search	Search	Search
2026-07-31 23:40:29	admin	2026-07-31 23:40:29	admin	0	cb5ff43f931e8b50b467f9f7dd03d63f
2026-08-01 00:02:03	admin	2026-08-01 00:02:03	admin	0	1cbf38ff931e8b50b467f9f7dd03d6ab
2026-07-31 23:30:57	admin	2026-07-31 23:34:05	admin	1	(empty)
2026-08-01 05:41:11	admin	2026-08-01 05:41:11	admin	0	1a1fb43f931e8b50b467f9f7dd03d6ba
2026-08-01 08:17:45	admin	2026-08-01 08:17:45	admin	0	1a1fb43f931e8b50b467f9f7dd03d6ba
2026-08-01 05:36:16	admin	2026-08-01 05:36:16	admin	0	1a1fb43f931e8b50b467f9f7dd03d6ba

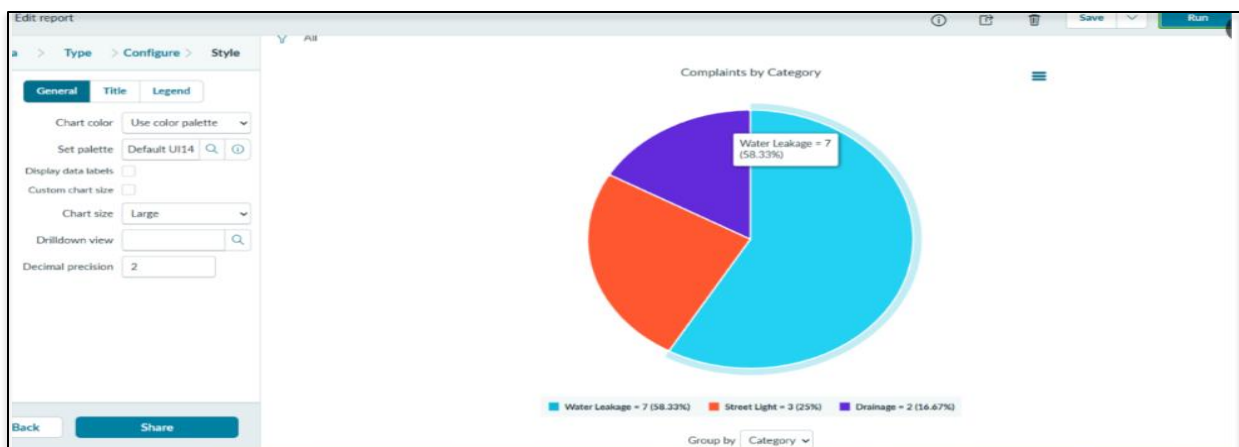
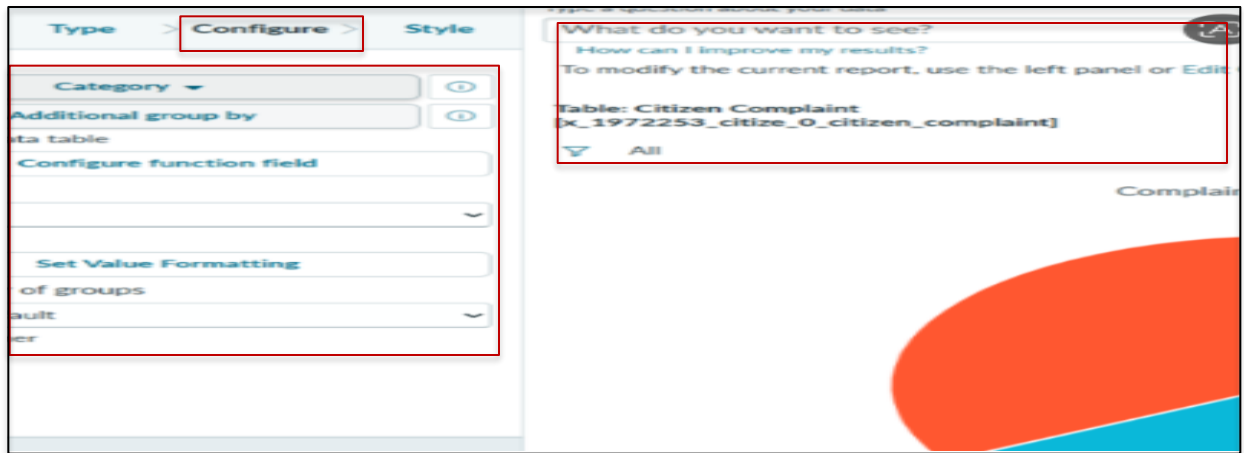
Activity: 4.2 Reports Validation

The following reports were tested:

- Complaints by Category
- Complaints by Status

- Complaints by Priority
- Complaints by Department
- AI Routed Complaints





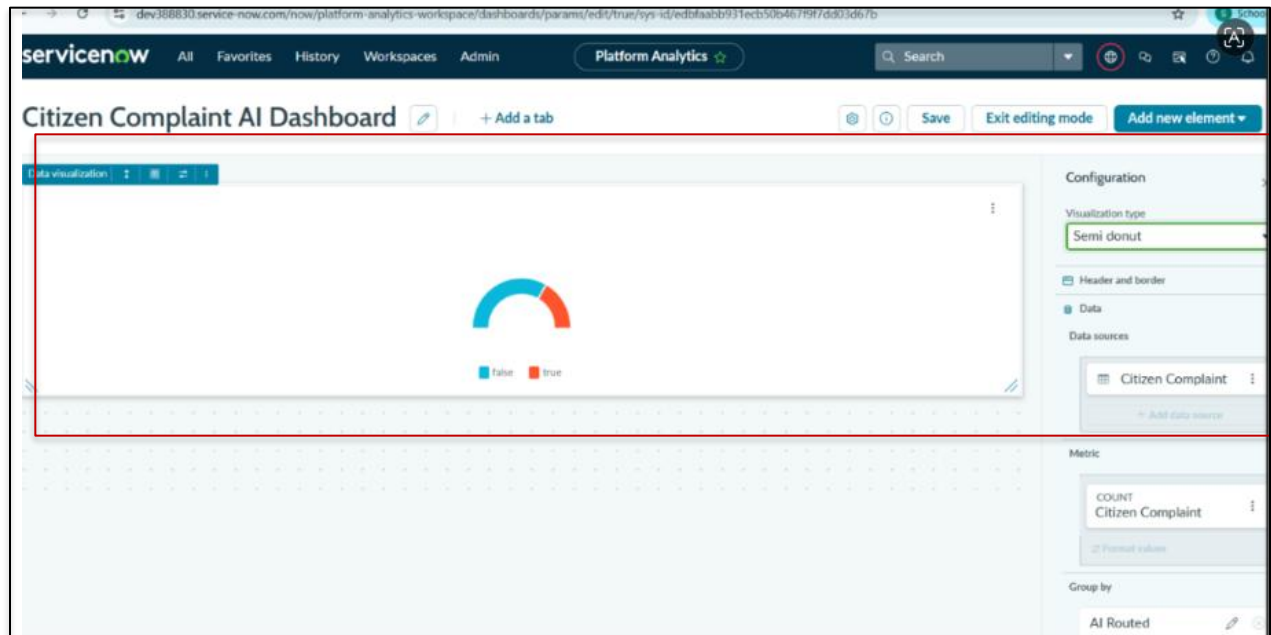
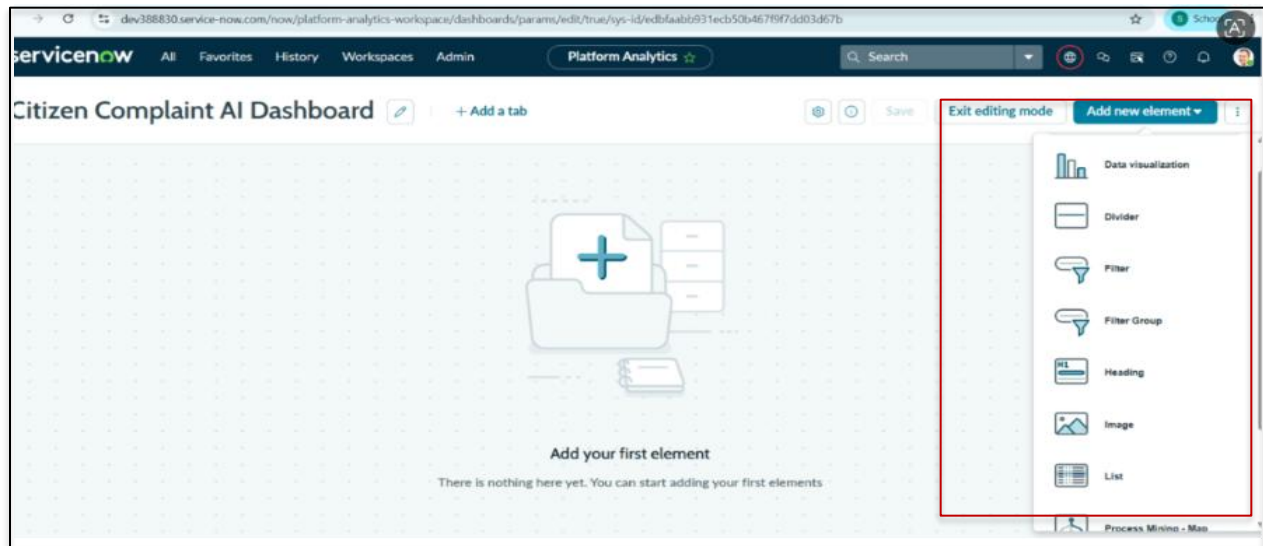
Activity: 4.3 Dashboard Validation

The Citizen Complaint AI Dashboard was validated.

Dashboard widgets display:

- Total Complaints
- Category-wise Complaints
- Department-wise Complaints
- Priority Distribution
- Complaint Status

Dashboard visualizations update automatically when new complaints are created.



Activity: 4.4 Knowledge Management Validation

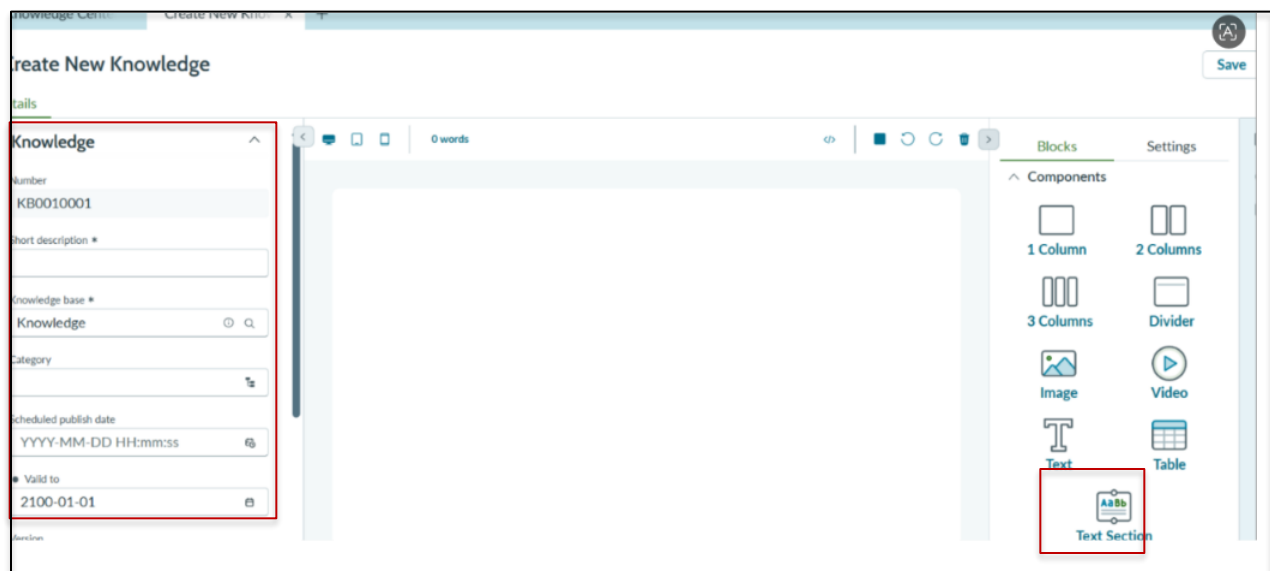
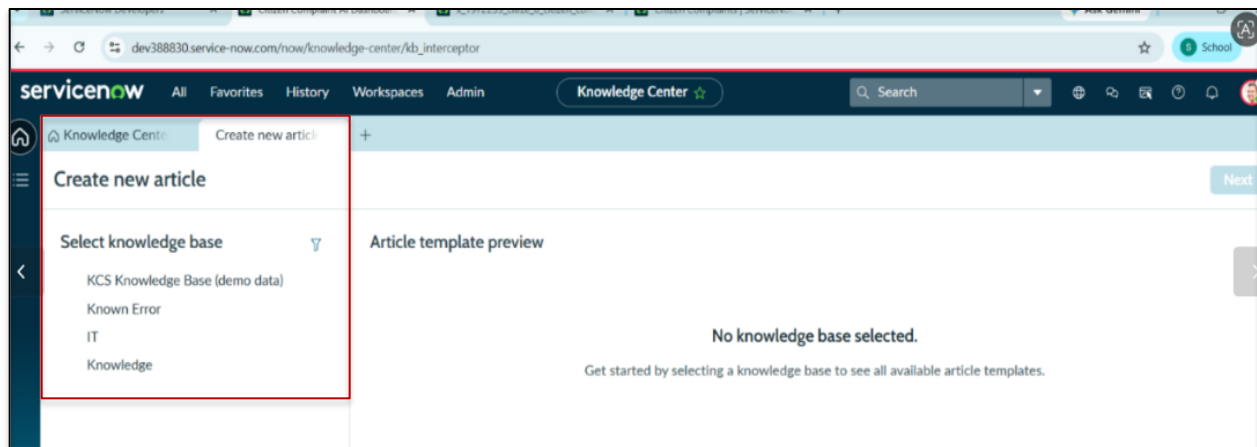
Knowledge Articles were tested.

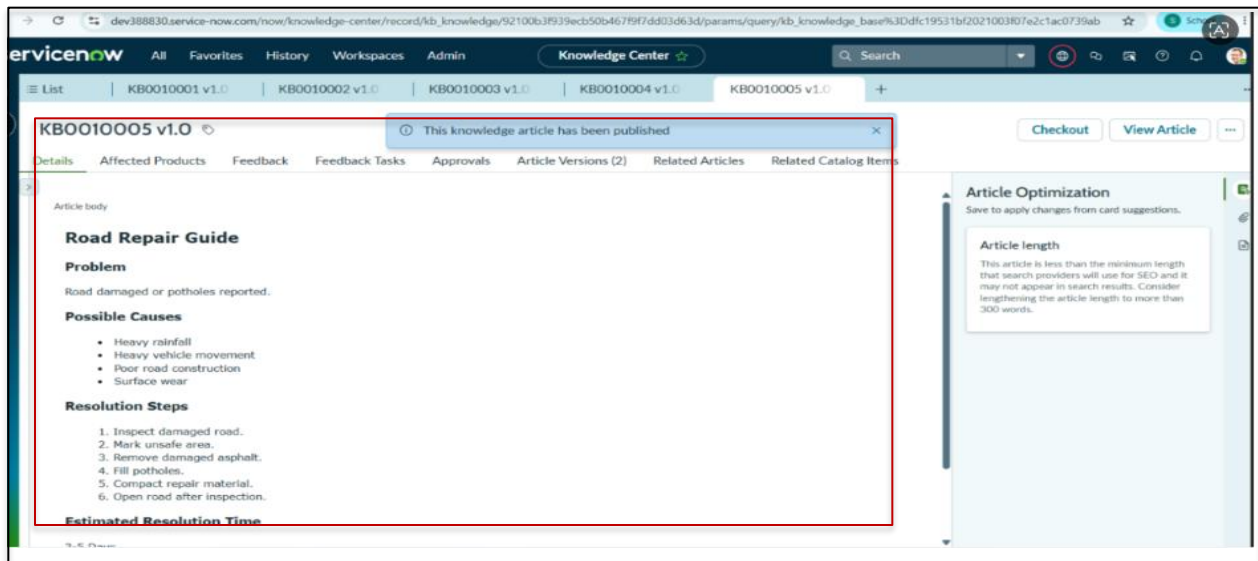
Articles created:

- Water Leakage Resolution
- Garbage Collection
- Street Light Maintenance
- Road Repair

- Drainage Blockage

Each article is accessible through the Knowledge Base.





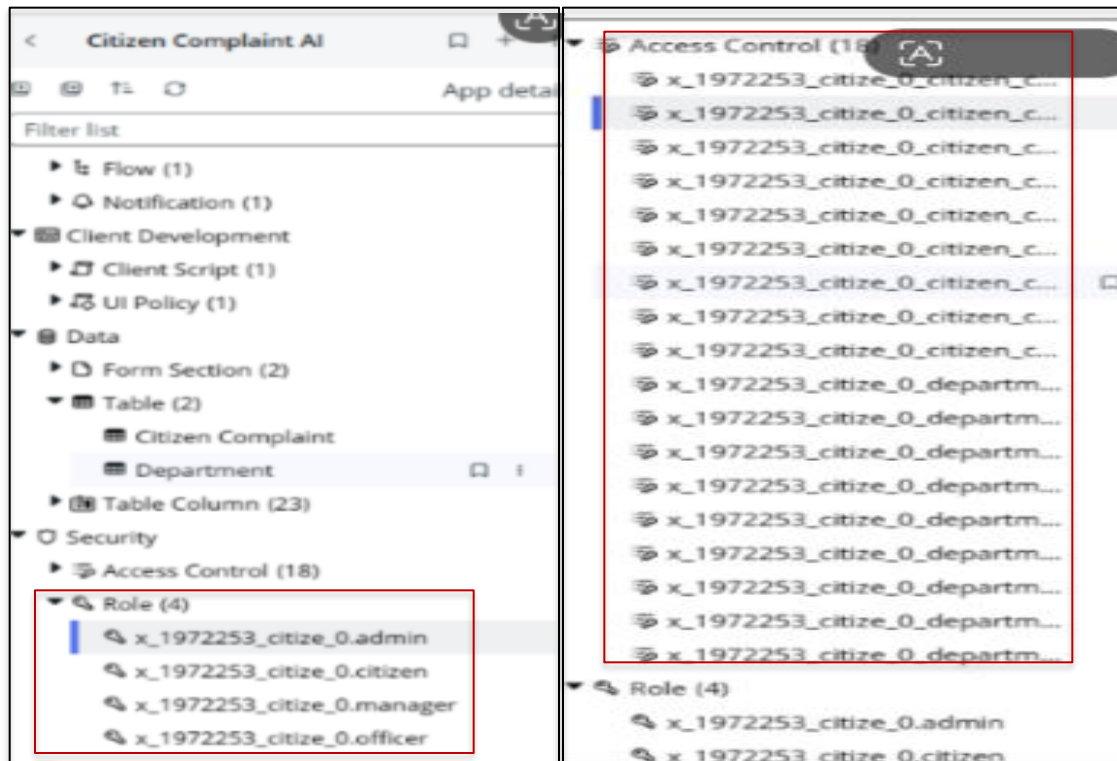
Activity: 4.6 Security & Access Control (ACLs)

Access Control Lists (ACLs) were implemented to secure application data.

Before creating ACLs, the Security Admin (Elevate Role) was enabled.

Roles Used

Role	Description
Citizen	Complaint Requester
Department Officer	Updates assigned complaints
Administrator	Full application management



PHASE 5 — Deployment & Final Presentation

This phase ensures that the **AI-Powered Citizen Complaint Management System** is successfully deployed, validated, and demonstrated as a complete ServiceNow application. It verifies that all implemented modules function correctly and are ready for end-user usage.

The application integrates complaint management, AI-assisted features, workflow automation, notifications, reporting, dashboards, and security into a centralized solution.

Activity: 5.1 Deployment Strategy

- Scoped Application Deployment
- Built using ServiceNow Studio (New Application Creation Model)

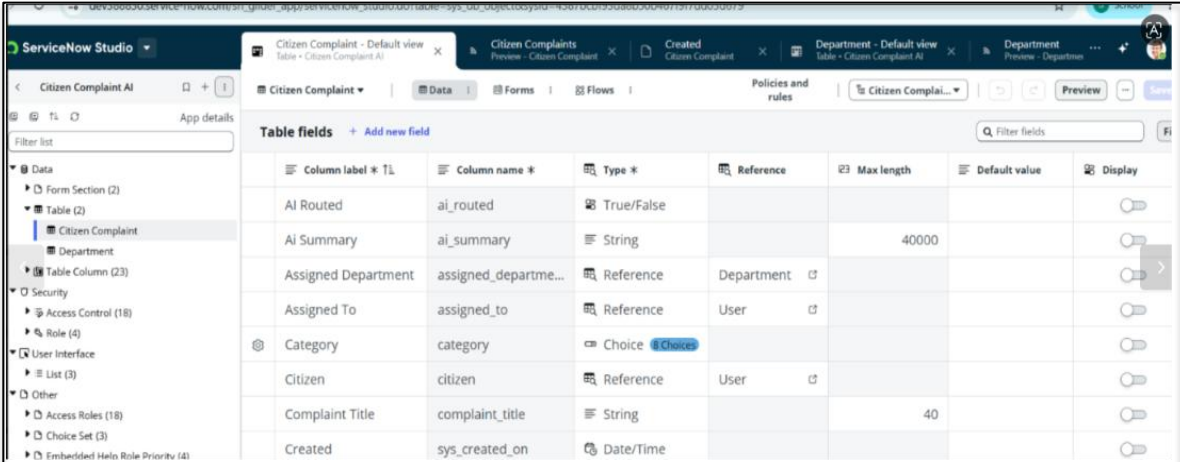
Deployment Environment Flow

1. Development Instance

The following configurations were completed:

- Scoped Application

- Custom Tables
- Forms
- Client Scripts
- UI Policies
- Business Rules
- Flow Designer
- Notifications
- ACLs
- Reports
- Dashboard
- Knowledge Articles
- AI Summary
- AI Recommendation
- AI Routing



The screenshot shows the ServiceNow Studio interface for configuring the 'Citizen Complaint' table. The 'Table fields' section is active, displaying a list of fields with their properties. The left sidebar shows the application structure, and the top navigation bar includes tabs for Data, Forms, Flows, and Policies and rules.

Column label * TL	Column name *	Type *	Reference	Max length	Default value	Display
AI Routed	ai_routed	True/False				☐
AI Summary	ai_summary	String		40000		☐
Assigned Department	assigned_departme...	Reference	Department			☐
Assigned To	assigned_to	Reference	User			☐
Category	category	Choice Choices				☐
Citizen	citizen	Reference	User			☐
Complaint Title	complaint_title	String		40		☐
Created	sys_created_on	Date/Time				☐

2. Testing Validation

The following functionalities were validated successfully:

- Complaint Registration

- Automatic Priority Assignment
- Automatic Department Assignment
- AI Summary Generation
- AI Recommendation Generation
- Notification Delivery
- Reports
- Dashboard
- Knowledge Articles
- ACL Security

3. Ready for Deployment

- The application is packaged as a **Scoped Application** and can be deployed to another ServiceNow instance using the **Application Repository** or **Update Sets**.

Activity: 5.2 Final End-to-End Demo Flow

Citizen Opens Application



Creates Complaint



Client Script Assigns Priority



Business Rule Assigns Department



AI Summary Generated



AI Recommendation Generated



Flow Designer Sends Notification



Department Resolves Complaint



Resolution Updated



Complaint Closed



Reports & Dashboard Updated

Activity: 5.3 AI Features Implemented

The application includes AI-inspired automation features developed using ServiceNow capabilities.

AI Summary

Generates a concise summary of the citizen complaint.

AI Recommendation

Suggests an appropriate resolution based on the complaint category.

AI Routing

Automatically assigns complaints to the appropriate municipal department.

Knowledge Management

Knowledge Articles provide standard operating procedures for common complaint categories

Activity: 5.4 Reports & Dashboard

The application includes multiple reports for monitoring complaint trends.

Reports Created

- Complaints by Category
- Complaints by Status
- Complaints by Priority
- Complaints by Department
- AI Routed Complaints

Dashboard

The **Citizen Complaint AI Dashboard** provides:

- Total Complaints
- Category-wise Distribution

- Department-wise Distribution
- Priority Analysis
- Status Analysis

The dashboard updates automatically whenever complaint records change.

Activity: 5.5 Key Learning Outcomes

By completing this project, you learned:

- Creatin a Scoped Application
- Creating Custom Tables
- Designing Forms
- Implementing Client Scripts
- Implementing UI Policies
- Developing Business Rules
- Automating processes using Flow Designer
- Configuring Notifications
- Implementing Role-Based Security using ACLs
- Creating Reports
- Designing Dashboards
- Managing Knowledge Articles
- Implementing AI-inspired features such as AI Summary, AI Recommendation, and AI Routing
- Understanding enterprise AI concepts such as AI Search, Now Assist, Virtual Agent, and Performance Analytics

Conclusion

The **AI-Powered Citizen Complaint Management System** successfully demonstrates the development of a complete end-to-end ServiceNow application using low-code and no-code capabilities. The system automates the entire complaint lifecycle, from complaint registration to resolution, while incorporating AI-inspired features such as AI Summary, AI Recommendation, and intelligent department routing. ServiceNow components including Flow Designer, Business Rules, Client Scripts, ACLs, reports, dashboards, and Knowledge Management work together to improve automation, security, transparency, and operational efficiency.

Although advanced enterprise AI features require additional licensing, the project effectively shows how AI concepts can be implemented using standard ServiceNow platform capabilities. Overall, the solution reduces manual effort, enhances citizen service delivery, and provides a scalable foundation for modern complaint management.